SAP ABAP CAPSTONE PROJECT

PROJECT TITLE:

CUSTOMER ORDER & PAYMENT TRACKING SYSTEM

USING SAP ABAP (O2C)

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1. INTRODUCTION

The Order-to-Cash (O2C) process is one of the core business processes in SAP Sales and Distribution (SD) module. It covers the complete lifecycle from customer order placement to payment receipt.

This project simulates the O2C process using SAP ABAP programming without relying on a real SAP system. Due to limited system access, internal tables are used to replicate SAP standard tables and transactions.

The system demonstrates how business transactions such as sales orders, billing, and payments are processed and analyzed using a custom ABAP report.

This project simulates SAP standard tables such as KNA1, VBAK, VBRK, and BSID using internal tables due to lack of system access.

2. PROBLEM STATEMENT

In real-world business scenarios, organizations face difficulty in tracking customer payments and identifying delays. Standard SAP reports can be complex and not always user-friendly for quick analysis.

There is a need for a simplified reporting system that can:

- Track customer transactions end-to-end
- Calculate payment delays
- Identify risky customers
- Provide clear and structured output

3. OBJECTIVE

The objective of this project is to:

- Simulate the complete O2C process
- Develop a custom ABAP report
- Analyze customer payment behavior
- Provide delay-based risk classification
- Demonstrate SAP-like system behavior without real database access

4. SOLUTION OVERVIEW

This project provides a simulated SAP O2C system using ABAP programming. It integrates customer, order, invoice, and payment data into a unified reporting system.

Key features include:

- End-to-end O2C process simulation
- Automated delay calculation
- Risk-based customer classification
- Interactive filtering system
- Structured report output similar to SAP ALV

5. SCOPE OF PROJECT

This project covers:

- Customer Master Data
- Sales Order Processing
- Invoice Generation
- Payment Recording
- Report Generation

The project is implemented using internal tables and modular ABAP programming.

6. O2C PROCESS FLOW

The following SAP transactions are simulated in this project:

Steps	Process	T-Code
1	Customer Creation	XD01
2	Sales Order Creation	VA01
3	Delivery Processing	VL01N
4	Billing (Invoice)	VF01
5	Payment Posting	F-28
6	Customer Line Items	FBL5N

The process flow:

Customer → Sales Order → Delivery → Invoice → Payment → Report

7. SYSTEM DESIGN

7.1 Data Simulation

Instead of real SAP tables, the following internal tables are used:

- Customer Table (simulates KNA1)
- Sales Order Table (simulates VBAK)
- Invoice Table (simulates VBRK)
- Payment Table (simulates BSID)

7.2 ABAP Program Structure

The program is divided into multiple sections:

- Type Definitions
- Internal Tables
- Data Loading
- Business Logic
- Filtering
- Output Display

8. FUNCTIONALITY

8.1 Data Linking

The program connects:

- Sales Orders → Invoices
- Invoices → Payments

8.2 Delay Calculation

Payment delay is calculated as:

Delay = Payment Date – Invoice Date

8.3 Risk Classification

Delay Days	Status
0-5	On Time
6-15	Late
>15	High Risk

8.4 User Filters

The report allows filtering by:

- Customer ID
- Payment Status
- Delayed Payments

8.5 Output Display

The report is displayed in a structured format similar to SAP ALV reports, showing:

- Customer details
- Order information
- Invoice data
- Payment status
- Delay days

9. TECHNOLOGIES USED

- SAP ABAP – Backend logic and report generation
- HTML/CSS – Frontend simulation interface
- VS Code – Development environment
- GitHub – Version control and project hosting

10. UNIQUE FEATURES

- Simulation of SAP system without real access
- End-to-end O2C process implementation
- Automated delay calculation
- Risk-based classification
- User-friendly report output
- Modular and clean ABAP code

11. FUTURE ENHANCEMENTS

- Integration with real SAP database
- ALV Grid with interactive features
- Graphical dashboard (charts)
- Email alerts for delayed payments
- Integration with SAP Fiori UI

12. CONCLUSION

This project successfully demonstrates the Order-to-Cash process using SAP ABAP in a simulated environment. It provides a clear understanding of business flow and technical implementation. The solution helps in identifying delayed payments and risky customers, which is crucial for business decision-making.

The project meets all requirements of an SAP capstone project. This project demonstrates both functional and technical understanding of SAP systems and can be extended to real-time enterprise applications.

13. SCREENSHOTS

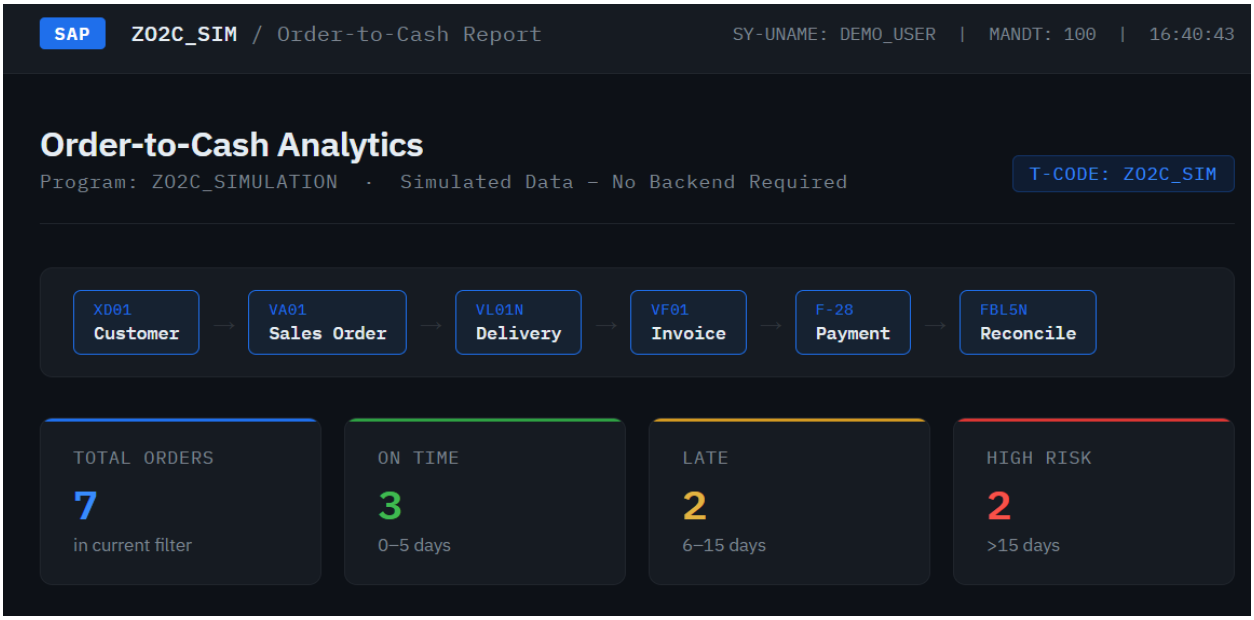

Figure 1: O2C Dashboard showing process flow

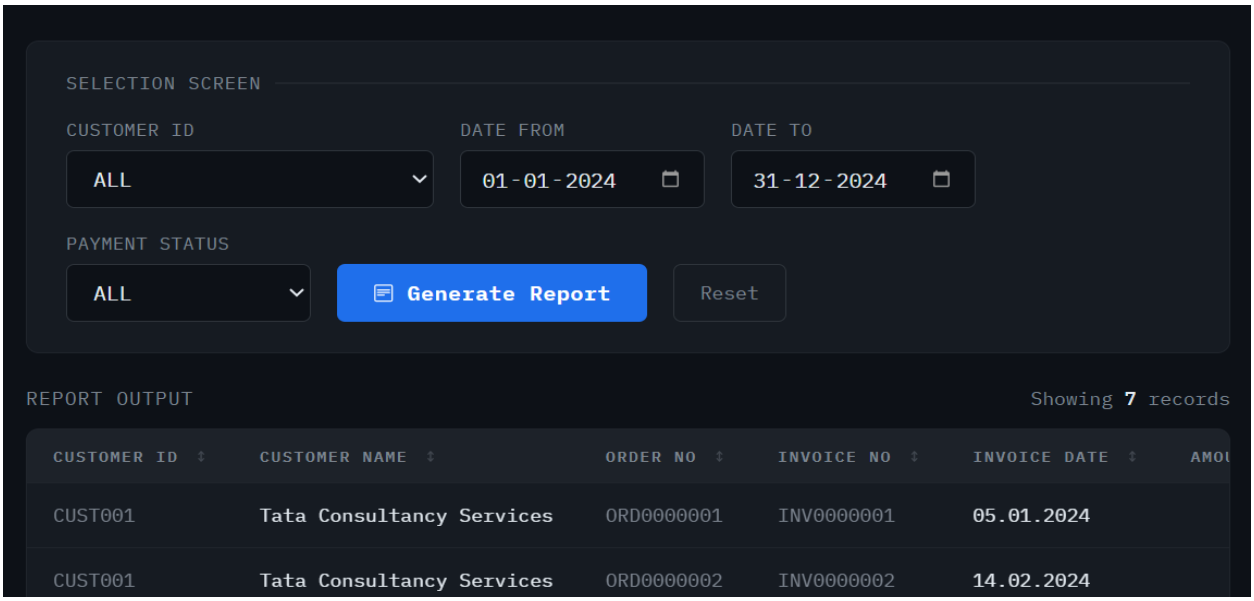

Figure 2: User input screen for filtering report data

CUSTOMER ID	CUSTOMER NAME	ORDER NO	INVOICE NO	INVOICE DATE	AMOU
CUST001	Tata Consultancy Services	ORD0000001	INV0000001	05.01.2024	
CUST001	Tata Consultancy Services	ORD0000002	INV0000002	14.02.2024	
CUST002	Infosys Limited	ORD0000003	INV0000003	18.01.2024	
CUST003	Wipro Technologies	ORD0000004	INV0000004	24.01.2024	
CUST004	HCL Technologies	ORD0000005	INV0000005	05.02.2024	
CUST004	HCL Technologies	ORD0000006	INV0000006	18.02.2024	
CUST005	Tech Mahindra Ltd	ORD0000007	INV0000007	10.02.2024	

ZO2C_SIMULATION · Simulated Data Only – No SAP DB Tables Used INR · MANDT 100 · ERP 6.0 EHP8

Figure 3: Generated O2C report displaying customer transactions and delay analysis